

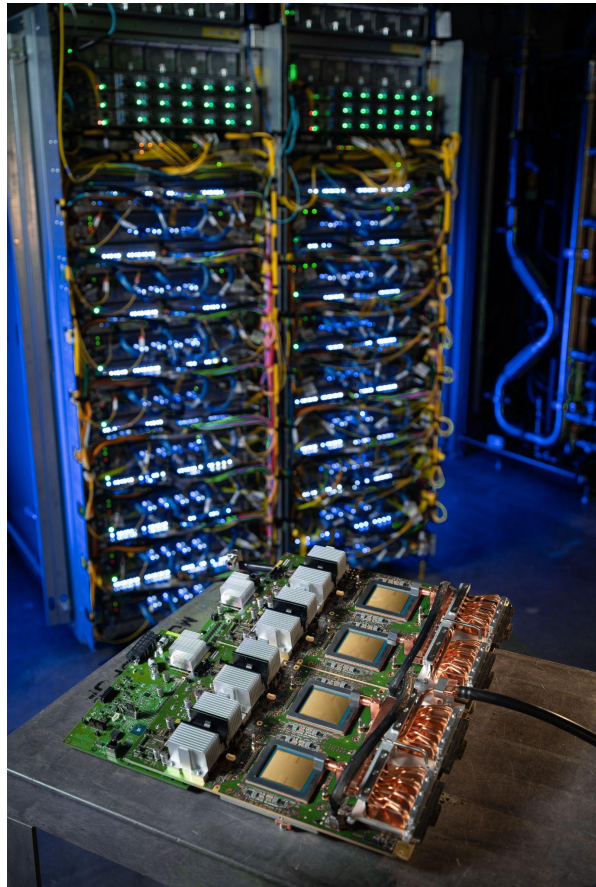


The Eighth Generation TPU Family: Two Chips Optimized for the Agentic Era

Norman P. Jouppi and Sridhar Lakshmanamurthy

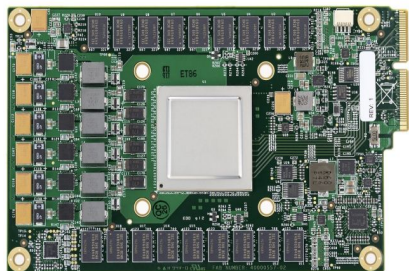
With contributions from many others

Hot Chips August 25, 2026



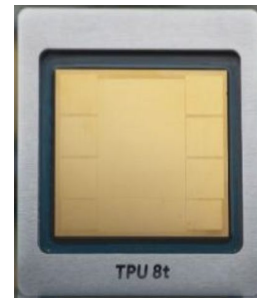
Google's Decade-Plus Custom TPU Strategy

- Since 2015 Google has been alternating between inference and training TPUs
 - The first TPU was a inference-only AI/ML accelerator
 - Then we deployed optimized training chips followed by a next generation optimized inference chip
 - We've iterated since then - for example: TPUv4i and TPUv4 (each published in ACM/IEEE ISCA)



2015 TPUv1
1 PCIe card
90 int8 TOPS

While increasing
shared-memory system
performance by
1,000,000X



2026 TPU8t
9600 chips per pod
120 FP4 EFLOPS

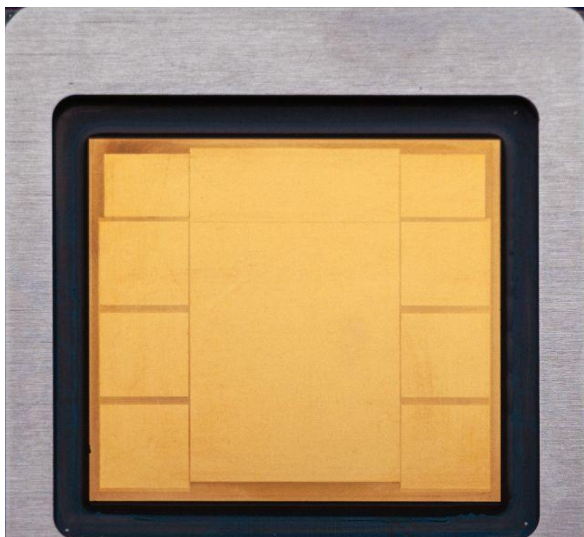
What's New for the 8th Generation TPUs

- Introducing new training and inference TPUs concurrently
 - TPU 8t (optimized for training)
 - TPU 8i (optimized for inference)
- More differentiation needed to support the diverse requirements of pre-training, distillation, sampling, Mixture-of-Experts (MoEs), and agentic models
 - Different interconnect topologies, 2X higher BW interconnects
- Built to handle long context windows and complex logic required by Agentic AI
 - With better connectivity to servers and high-performance storage
- Uniquely balanced solutions to optimize the entire ML pipeline
 - From large scale pretraining to agentic serving

Audience Participation Pop Quiz

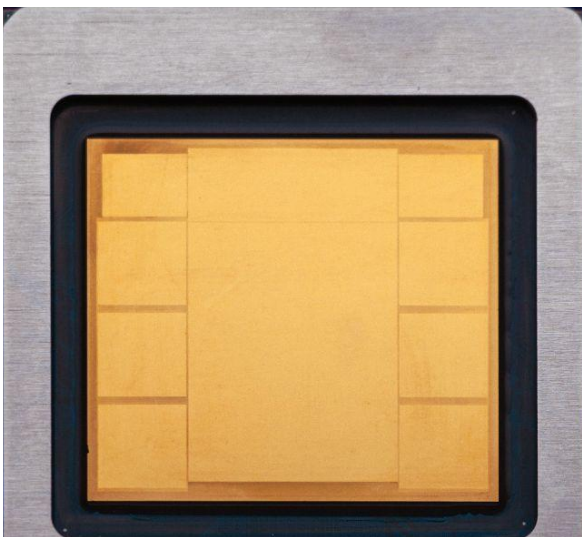
- Which chip is which?
 - Can you tell from their pictures?
 - There is a visual clue

Audience Participation Pop Quiz



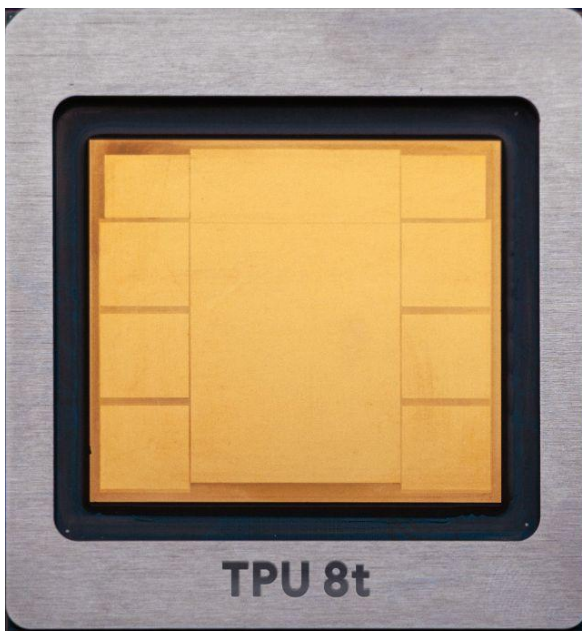
Audience Participation Pop Quiz

- Clue: Inference needs more HBM BW per compute



Audience Participation Pop Quiz

- Clue: Inference needs more HBM BW per compute



TPU 8i: The Low-latency Inference Powerhouse

Zero-wait silicon optimized for reasoning and RL sampling

Large SRAM

384MB

2.4x vs prior generation

HBM Capacity

288GB

12-Hi HBM3E stacks

ICI Bandwidth

19.2Tb/sec

2x vs. prior generation

Metric	SRAM (On-Chip)	HBM (Off-Chip)	Multiplier SRAM Advantage
Aggregate Bandwidth	~150–200 TB/s	~10–15 TB/s	15x – 20x
Access Latency	~1–2 ns	~150–200 ns	~100x
Energy per Bit (pJ/bit)	~0.1–0.5	~2.0–5.0	10x – 20x more efficient

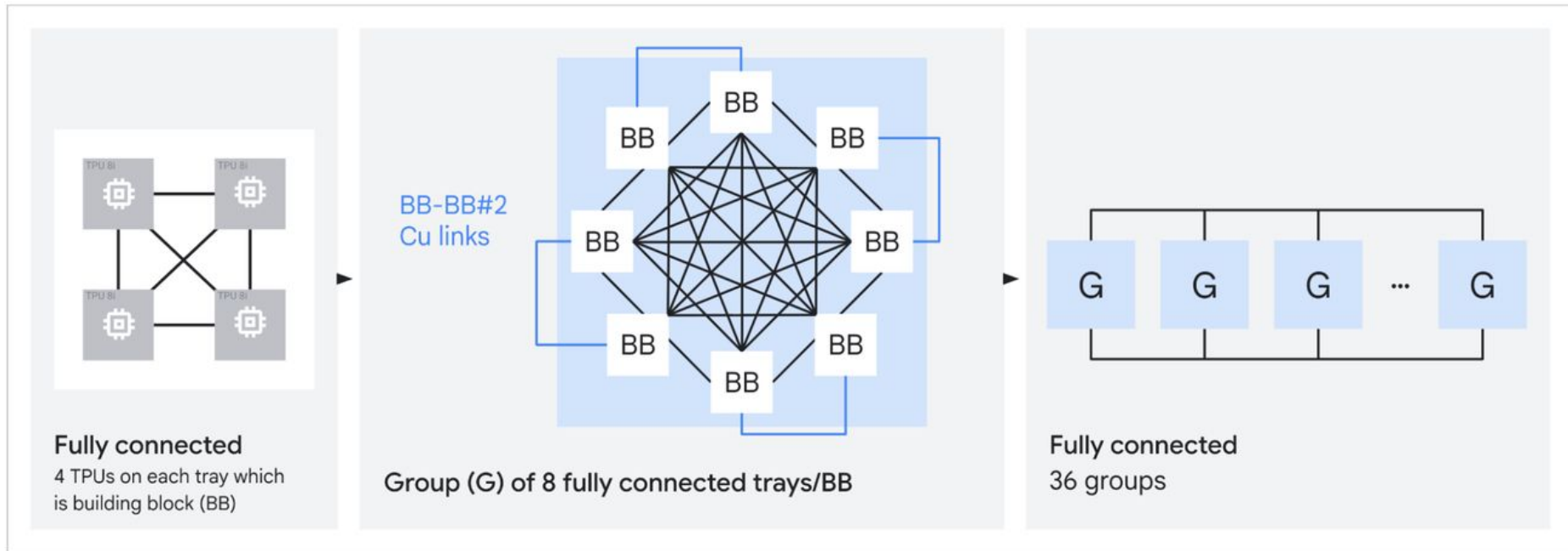
More SRAM

Breaks the memory wall

**More memory,
more bandwidth**

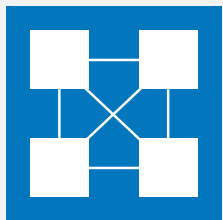
Larger models, more cache

TPU 8i: BoardFly Topology



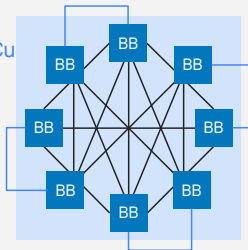
TPU 8i: Boardfly Network

Boardfly

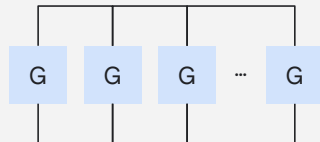


Fully connected
4 TPUs on each tray

BB-BB#2 Cu links



Group of 8 fully connected trays



Fully connected
Up to 36 groups

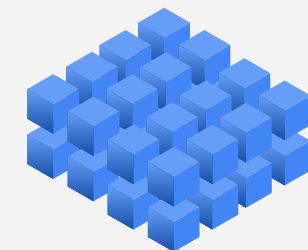
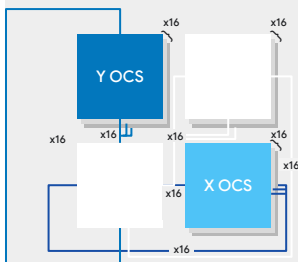
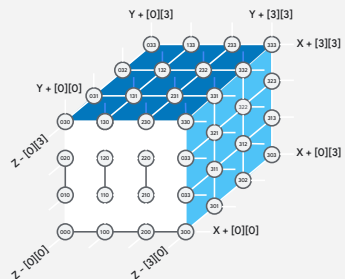
All-to-all in MoE inference
bound by network latency

Boardfly:

- 36 groups fully connected for a total of 1152 chips per pod
- Max 7 hops

Much lower #hops compared to 3D
Torus (e.g. 16)

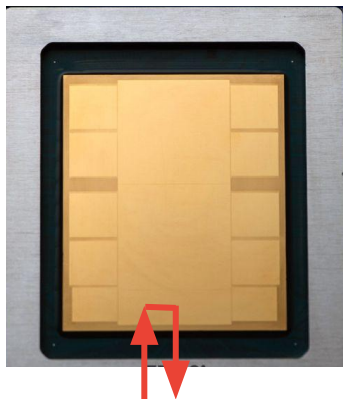
3D Torus



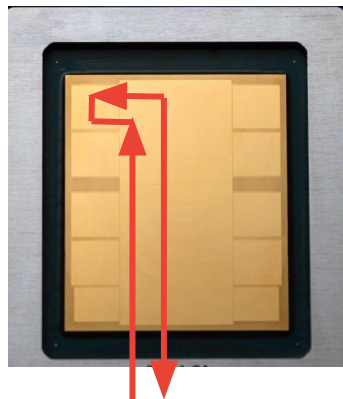
8x8x16

The Collective Acceleration Engine (CAE) Reduces Latency

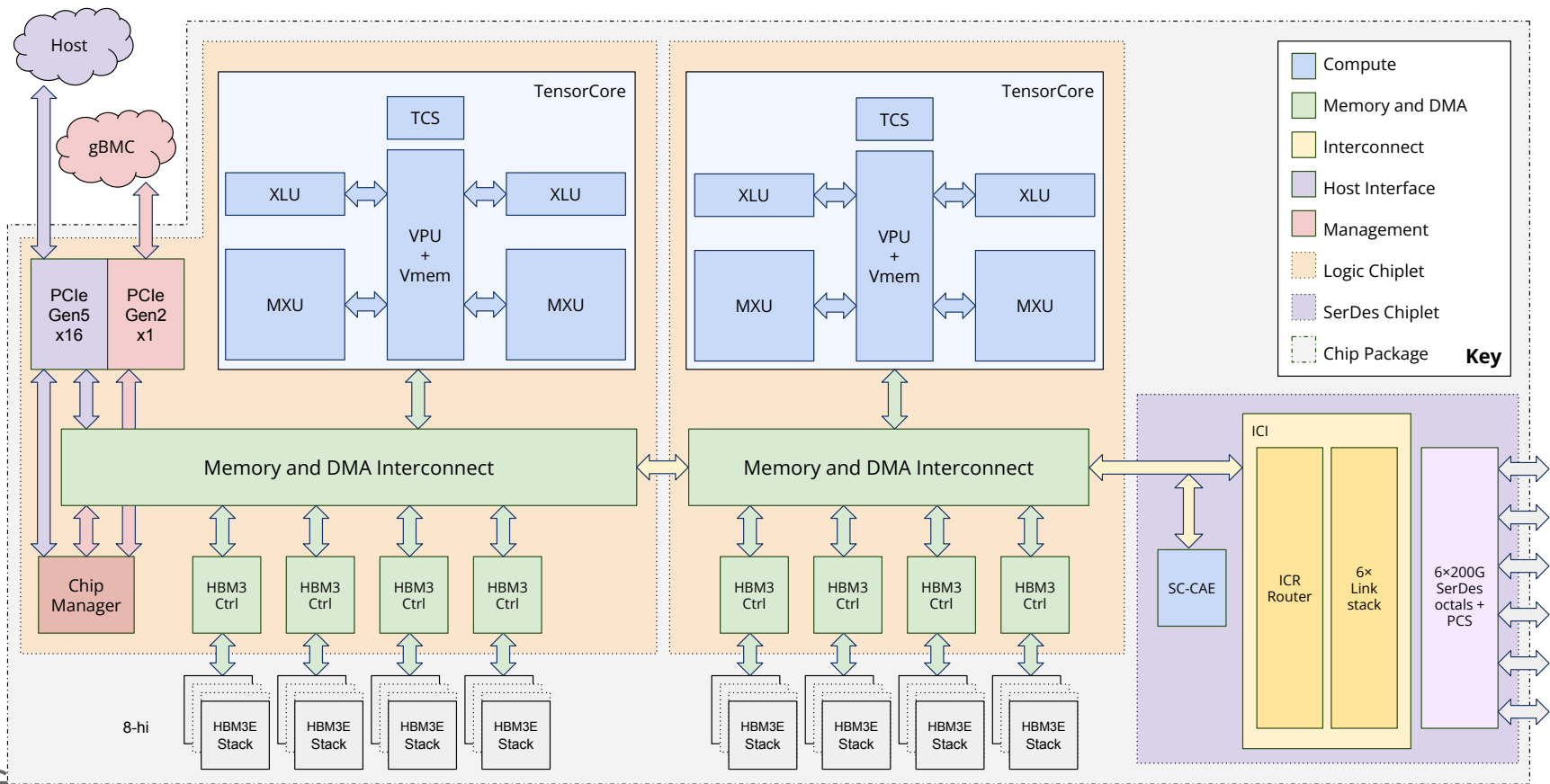
- **Efficiently performs in-network collectives**
 - Located in ICI I/O die
 - Historically in general networks this has been very complicated
- **Reduces on-chip latency by 5X:**
 - Reducing distance traveled within the package
 - Avoiding HBM access



Vs.



8i Block Diagram

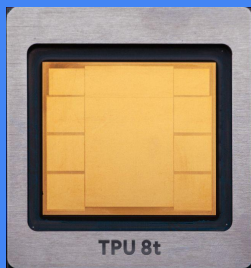


TPU 8t: The Training Powerhouse

Purpose-built for trillion-parameter scale.

8t Superpod

2.4 TBps I/O BW per chip for glueless scale-up



2PB Memory

Total Shared HBM

9,600 Chips

Single Superpod Scale

121 Exaflops

3x over Ironwood

Native FP4 Support

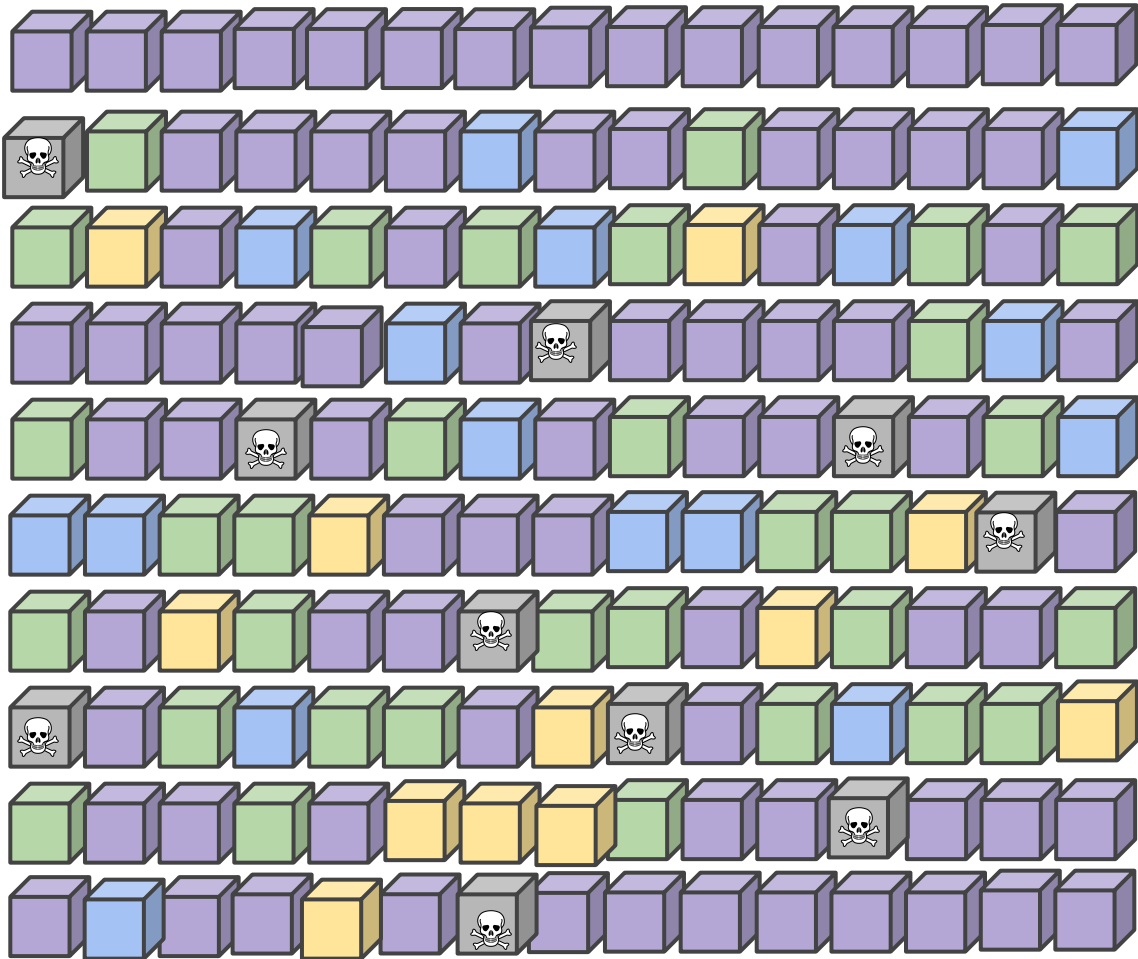
Enables large scale lower precision workloads

2x Perf-per-Watt vs. Ironwood

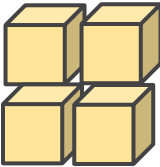
Enhanced power-to-goodput ratio for sustainable AI.

9600 TPU 8t Chips Sharing Memory Using OCS

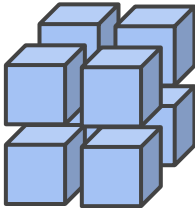
Shared-memory Superpod



Any slice size or shape can be created, limited only by pool size and availability



8x8x4



8x8x8

Scale Without Borders: The Virgo Network

A global non-blocking cluster fabric for the Agentic Era.

134,400 TPUs

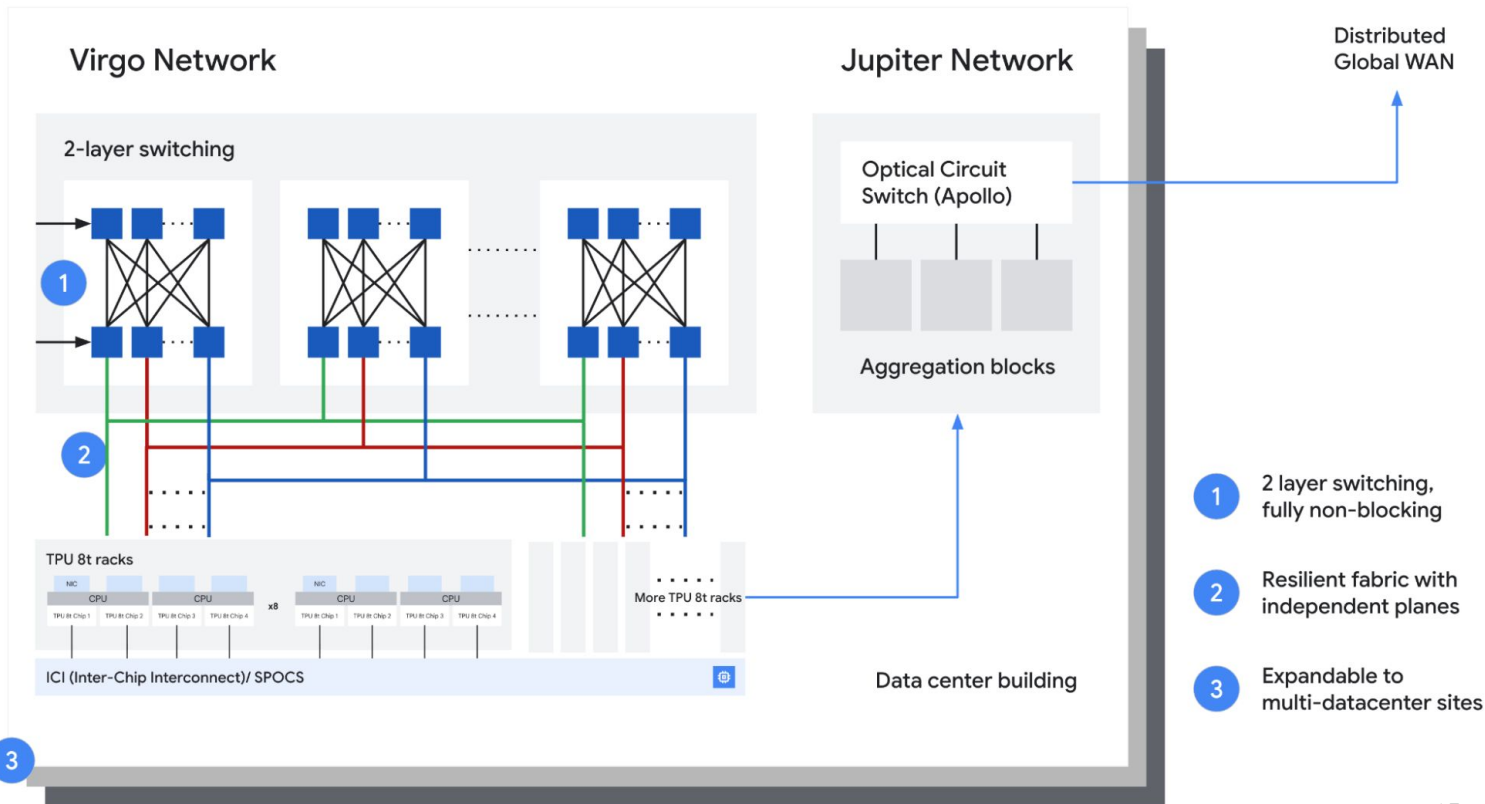
In a single Virgo cluster

1.6 YottaFlops

Peak compute power

47

Petabits/s

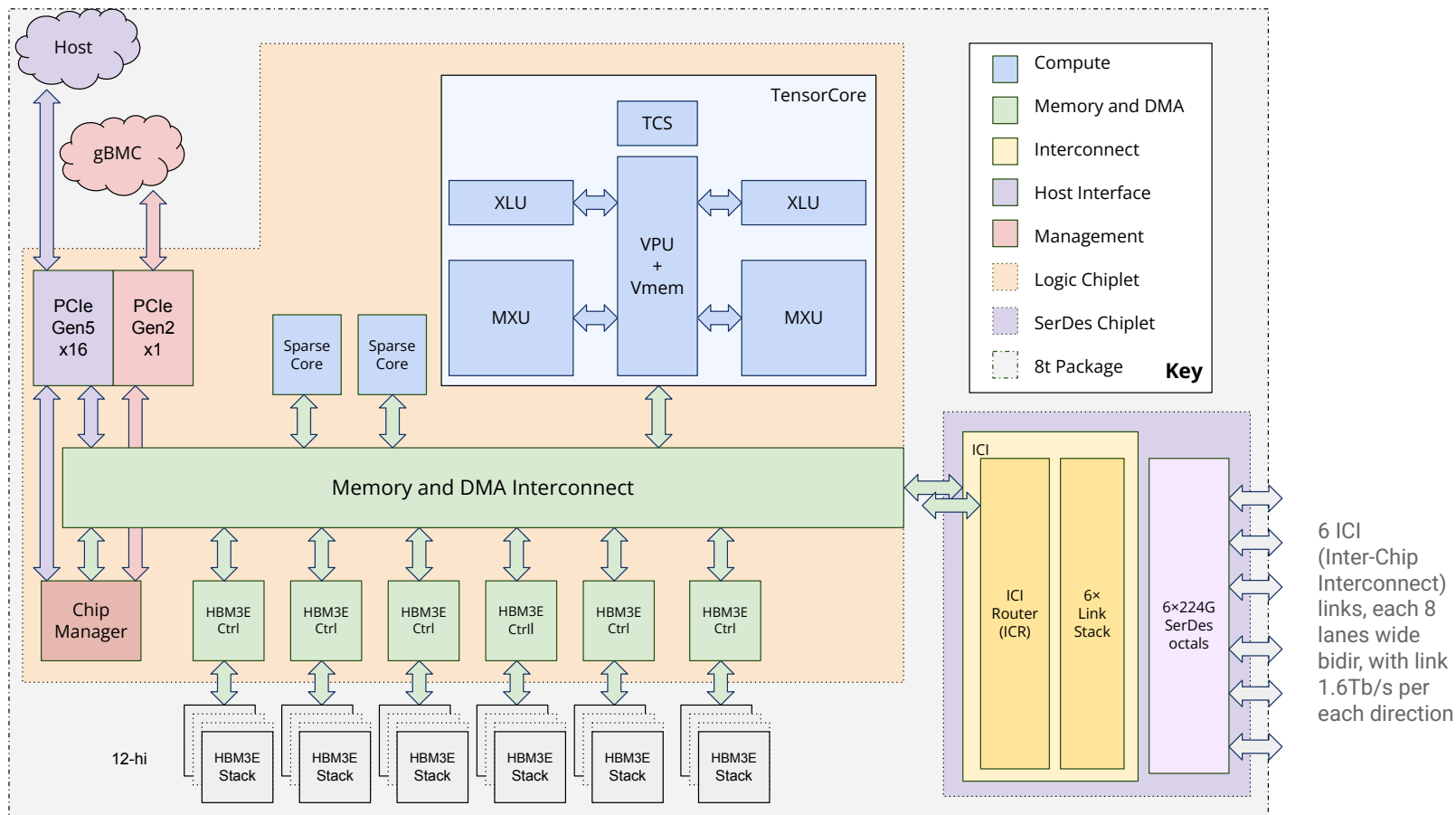


Reliability and Availability is a Priority

Critical to sustain near 100% goodput on 100K TPU systems

- Enhanced HBM link protection via CRC
- Retry on HBM UECC and D/Q parity errors, link errors
- Additional on-chip interconnect control path parity protection
- Improved sensors (V, T, droop, aging) and telemetry
- In-field unit test during idle cycles
- Advanced fleet health monitoring
- Superior water cooling
 - Failure rates increase exponentially with temperature
 - **Arrhenius equation:** Chemical degradation and semiconductor wear roughly double for every 10 °C to 15 °C increase in operating temperature

8t Block Diagram

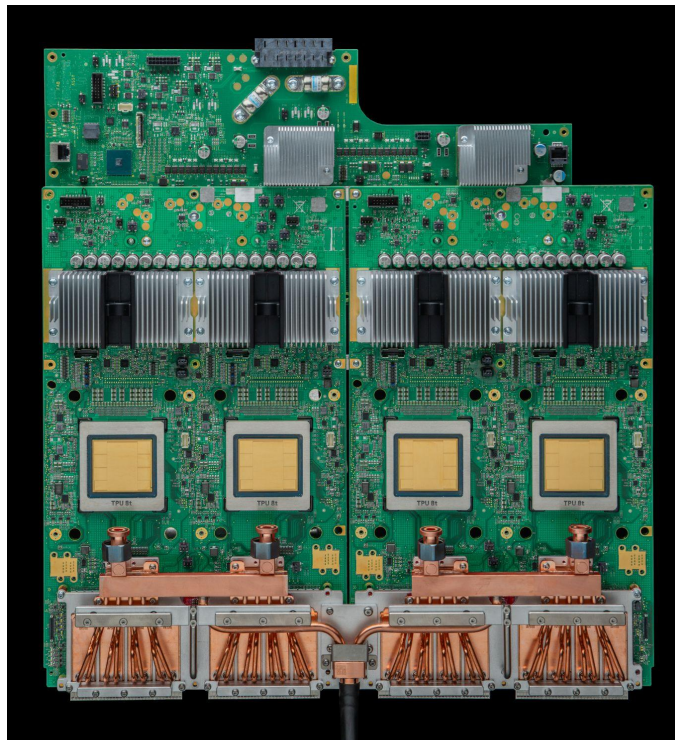


TPU 8t Tray

First stage voltage regulators →

TPU 8t packages →

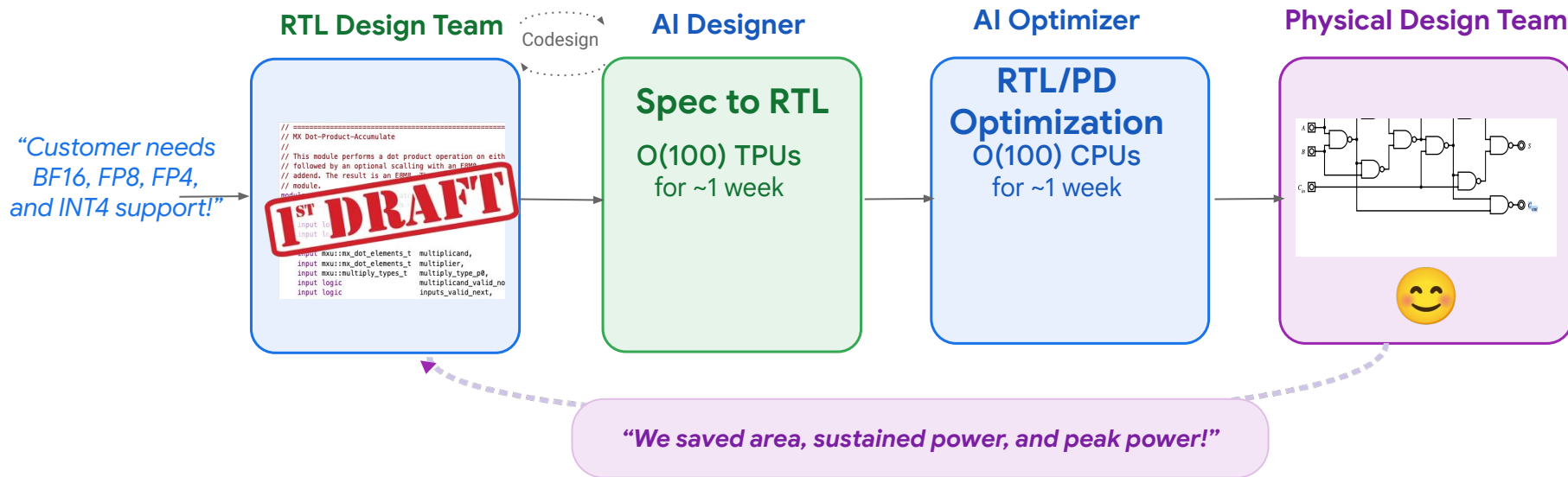
Water cooled optics
(an ML first) →



The 8t Rack (6 out of 300 in superpod)



Using AI/TPUs to Build the next AI/TPUs



TPU 8t

- **TensorCore (MXU):**
 - 6% power reduction
 - 5.8% area reduction => 6% additional TFLOPs
- **SparseCore :**
 - 13% area reduction => 10% additional TFLOPs

TPU 8i

- **TensorCore (MXU):** 5.3% MXU area reduction
 - Realized 5% more FLOPs in the same thermal budget.
- **SparseCore:** Helped close aggressive timing targets

Software and Developer Experience

- **Pallas and Mosaic:**

- Leverage our custom kernel language, Pallas, to write hardware-aware kernels directly in Python.
- Allows for maximum performance optimization on the TPU 8i CAE and TPU 8t SparseCore.

- **Native PyTorch Integration:**

- Transitioning to TPUs is seamless for PyTorch users.
- Run existing models with full support for native features like Eager Mode

- **Seamless Portability:**

- Your existing JAX, PyTorch, or Keras codebases are fully compatible with this generation.
- Accelerated Linear Algebra (XLA) manages the complexities of Boardfly topology and CAE synchronization automatically
- Users focus on their models rather than the underlying TPU and interconnect architecture.

Summary

- We've been deploying accelerators optimized for inference and training for a decade
 - We've optimized for both training and inference while maintaining architectural compatibility
- We significantly improve perf/TCO, perf/W, latency, and throughput
 - Even without Moore's Law scaling and Dennard scaling
 - > 2X perf/TCO and 2X perf/W generation over generation
- This is due to the ML-first design of our architecture
 - Codesigned with compiler writers, enabling compiler optimizations that are not possible with fixed kernels
 - John Hennessy: "Never put off until runtime what you can do at compile time"
 - We've made this a foundational principle for TPUs
- This has enabled superior perf/TCO, perf/W, scalability, and RAS

